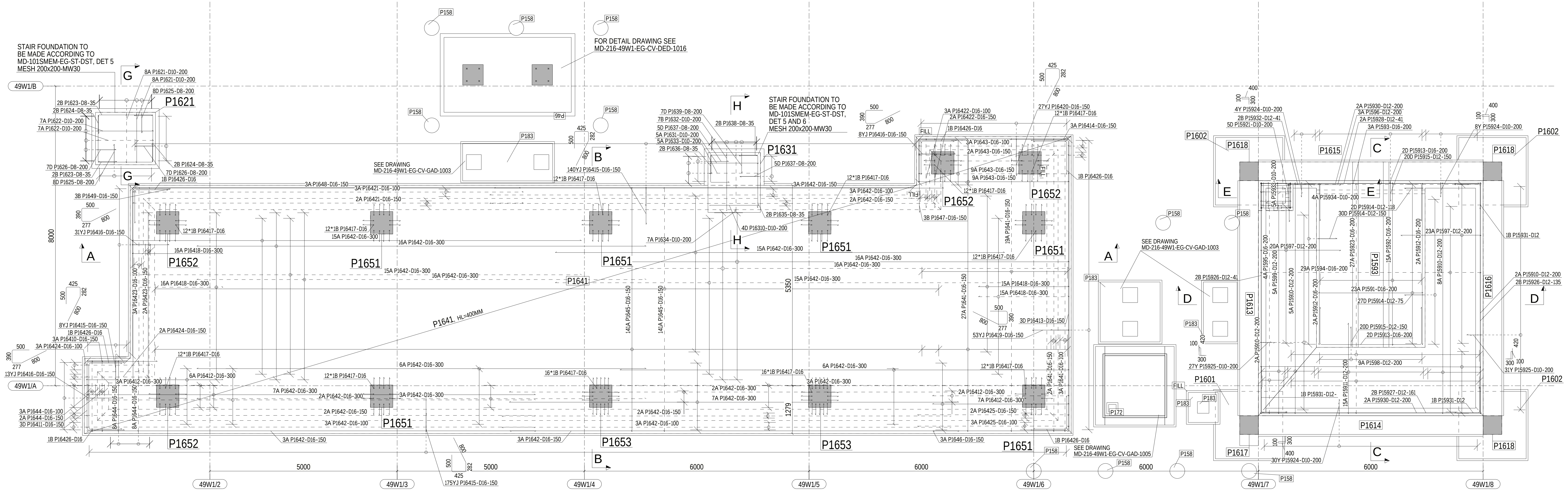
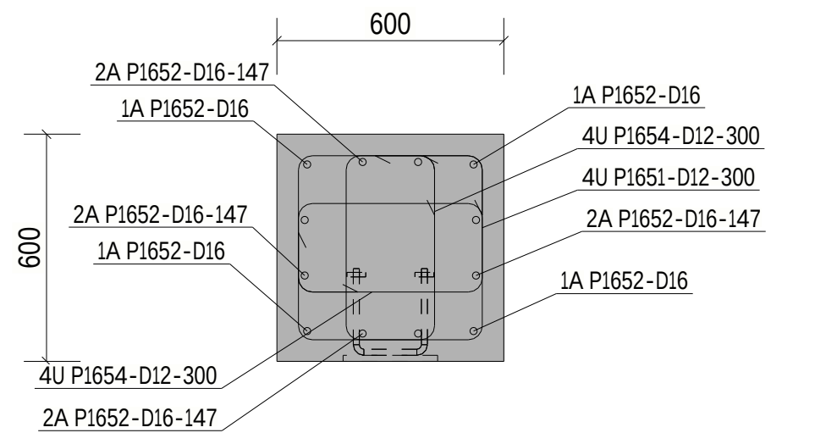
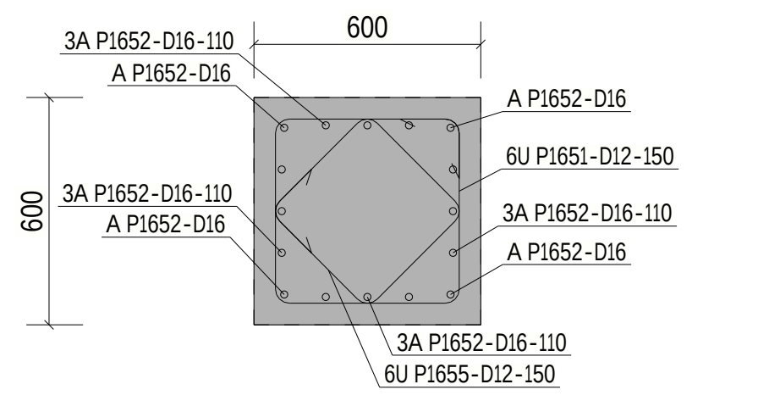
MD-216-49W1-EG-CV-DED-1021
DRAWING NUMBER
REV D01


STAIR FOUNDATION TO BE MADE ACCORDING TO MD-101SMEM-EG-ST-DST, DET 5 MESH 200x200-MW30
STAIR FOUNDATION TO BE MADE ACCORDING TO MD-101SMEM-EG-ST-DST, DET 5 AND 6 MESH 200x200-MW30
FOR DETAIL DRAWING SEE MD-216-49W1-EG-CV-DED-1016
SEE DRAWING MD-216-49W1-EG-CV-GAD-1003
SEE DRAWING MD-216-49W1-EG-CV-GAD-1005
P1621
P1631
P1641
P1651
P1652
P1601
P1602
P1613
P1614
P1615
P1616
P1617
P1618
P158
P159
P172
P181
P183
49W1/A
49W1/B
49W1/2
49W1/3
49W1/4
49W1/5
49W1/6
49W1/7
49W1/8
5000
6000
8000
5350
1279

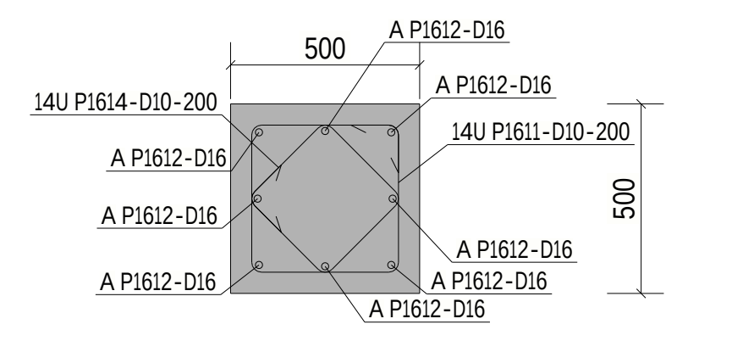
R.C.C GROUND SLAB REINFORCEMENT
(SCALE 1:50)


600
600
2A P1652-D16-147
1A P1652-D16
4U P1654-D12-300
4U P1651-D12-300
2A P1652-D16-147

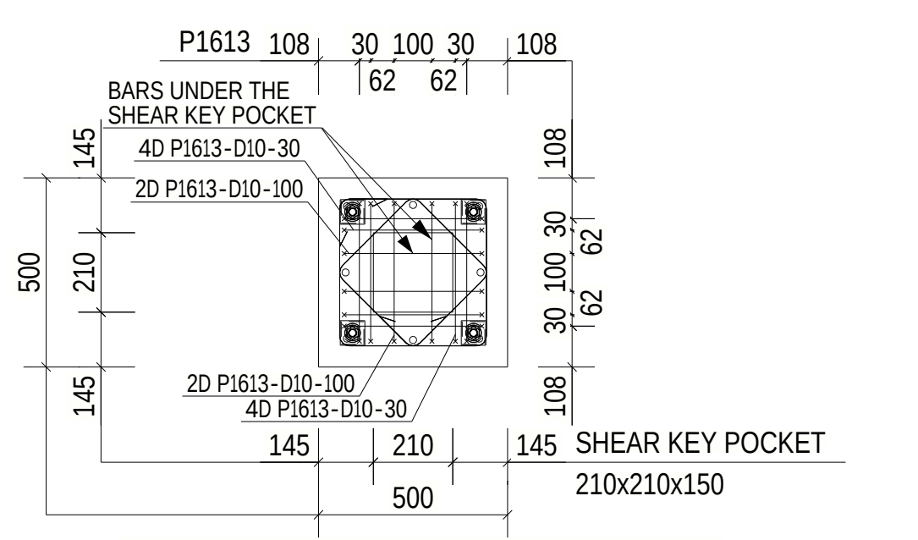
R.C. DET OF COLUMN P165-1 / P165-2
(SCALE 1:20)


600
600
3A P1652-D16-110
6U P1651-D12-150
3A P1652-D16-110
6U P1655-D12-150

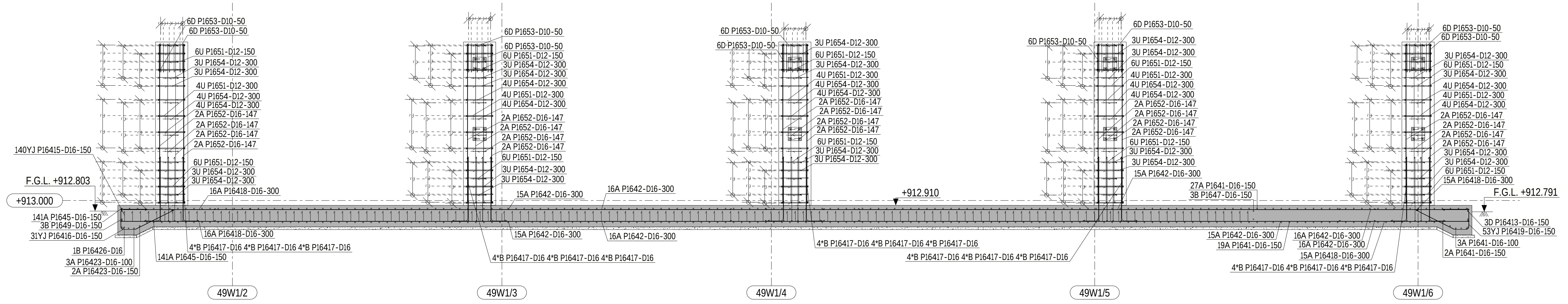
R.C. DET OF COLUMN P165-3
(SCALE 1:20)


500
500
14U P1614-D10-200
14U P1611-D10-200
A P1612-D16
A P1612-D16

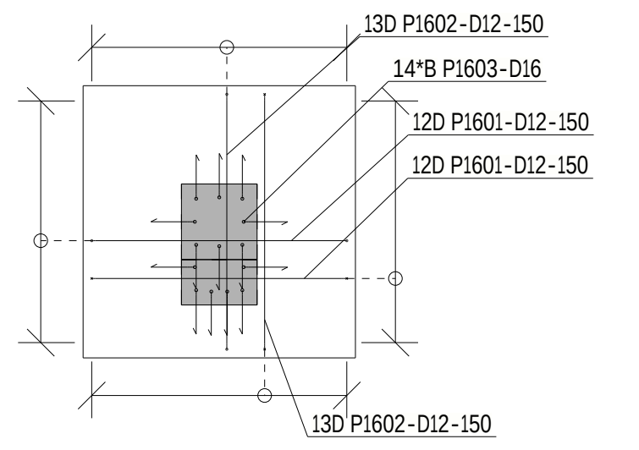
R.C DET. OF COLUMN P161-7 / P161-8
(SCALE 1:20)


P1613
BARS UNDER THE SHEAR KEY POCKET
40 P1613-D10-30
20 P1613-D10-100
SHEAR KEY POCKET 210x210x150
500
500
145
210
145

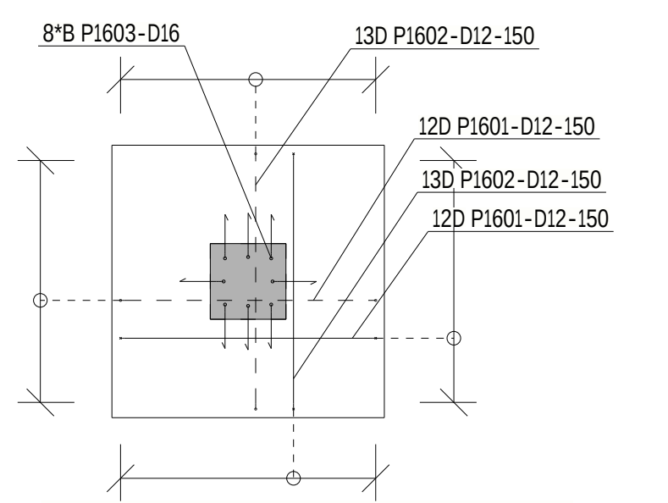
R.C DET. OF TOP OF COLUMN P161-7 / P161-8
(SCALE 1:20)


F.G.L. +912.803
+913.000
+912.910
F.G.L. +912.791
49W1/2
49W1/3
49W1/4
49W1/5
49W1/6

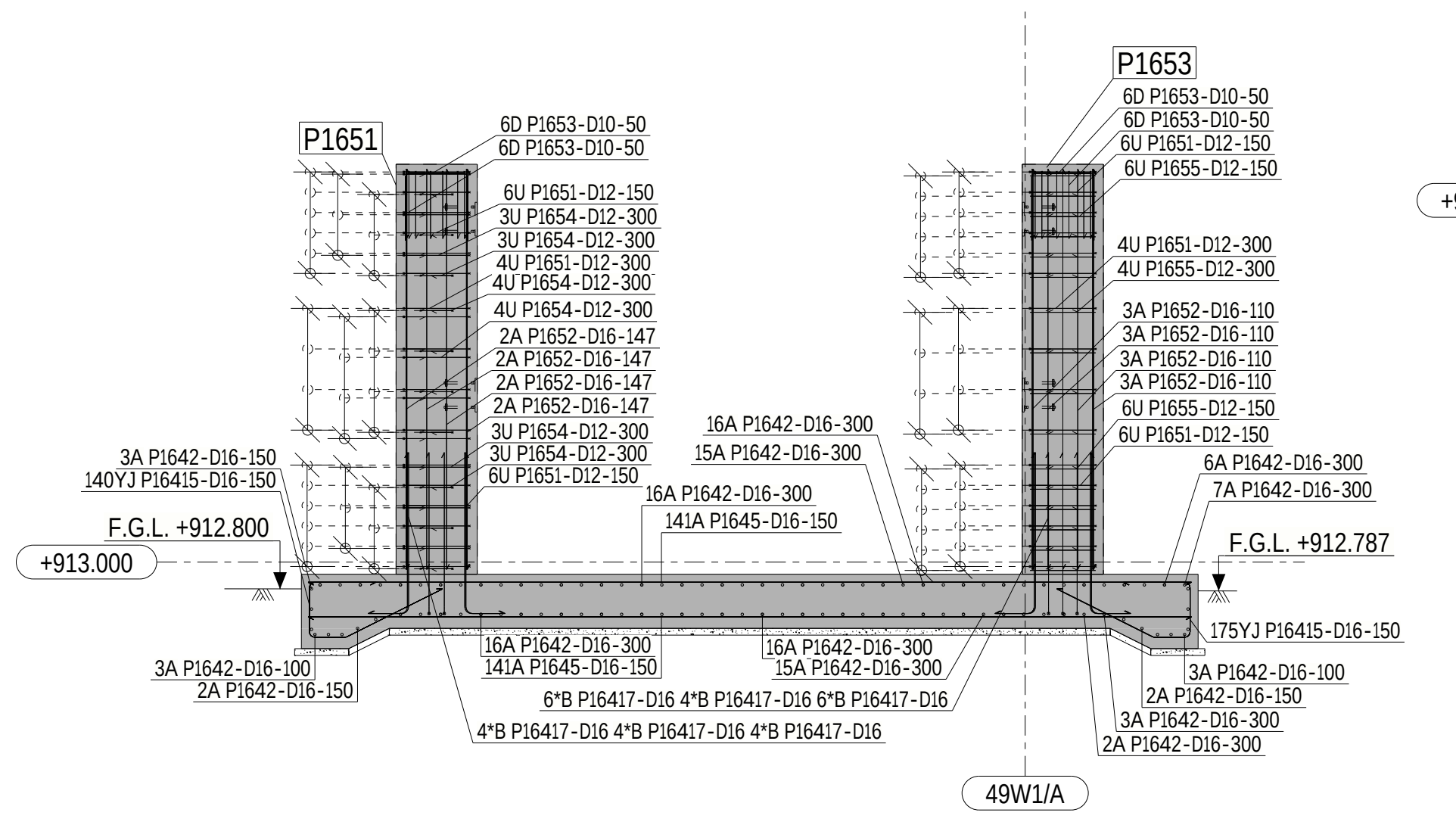
SECTION : A - A
(SCALE 1:50)


13D P1602-D12-150
14*B P1603-D16
12D P1601-D12-150
12D P1601-D12-150
13D P1602-D12-150

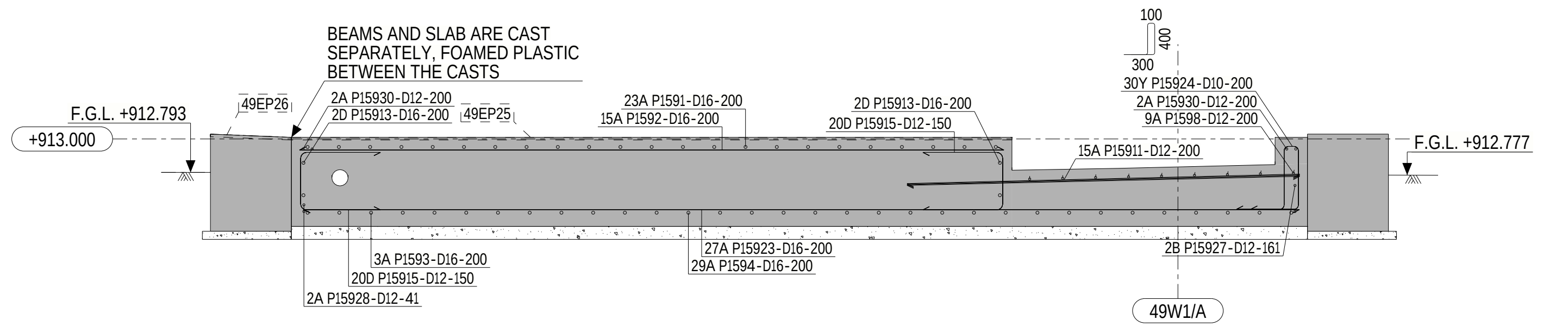
RC. DET. OF FOUNDATION P1601
(SCALE 1:50)


8*B P1603-D16
13D P1602-D12-150
12D P1601-D12-150
13D P1602-D12-150
12D P1601-D12-150

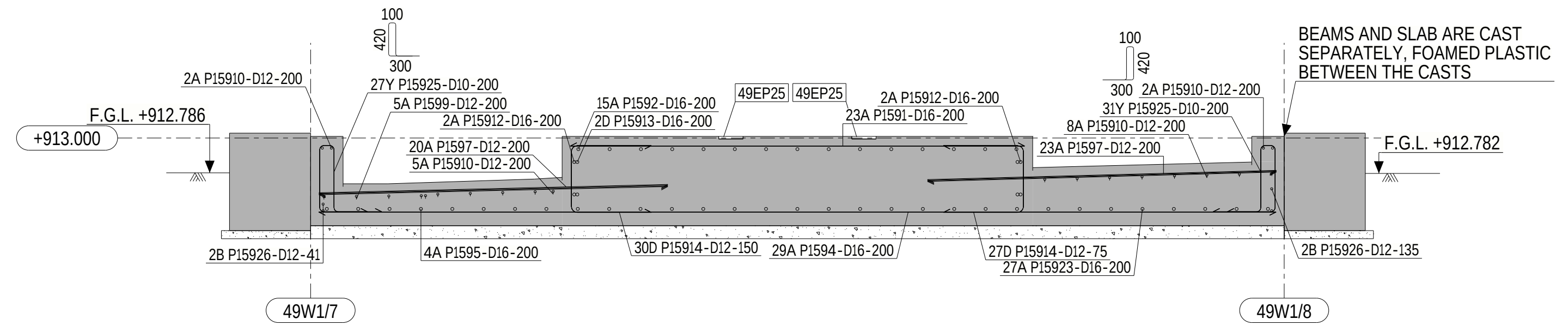
RC. DET. OF FOUNDATION P1602
(SCALE 1:50)


P1651
P1653
F.G.L. +912.800
+913.000
F.G.L. +912.787
49W1/A

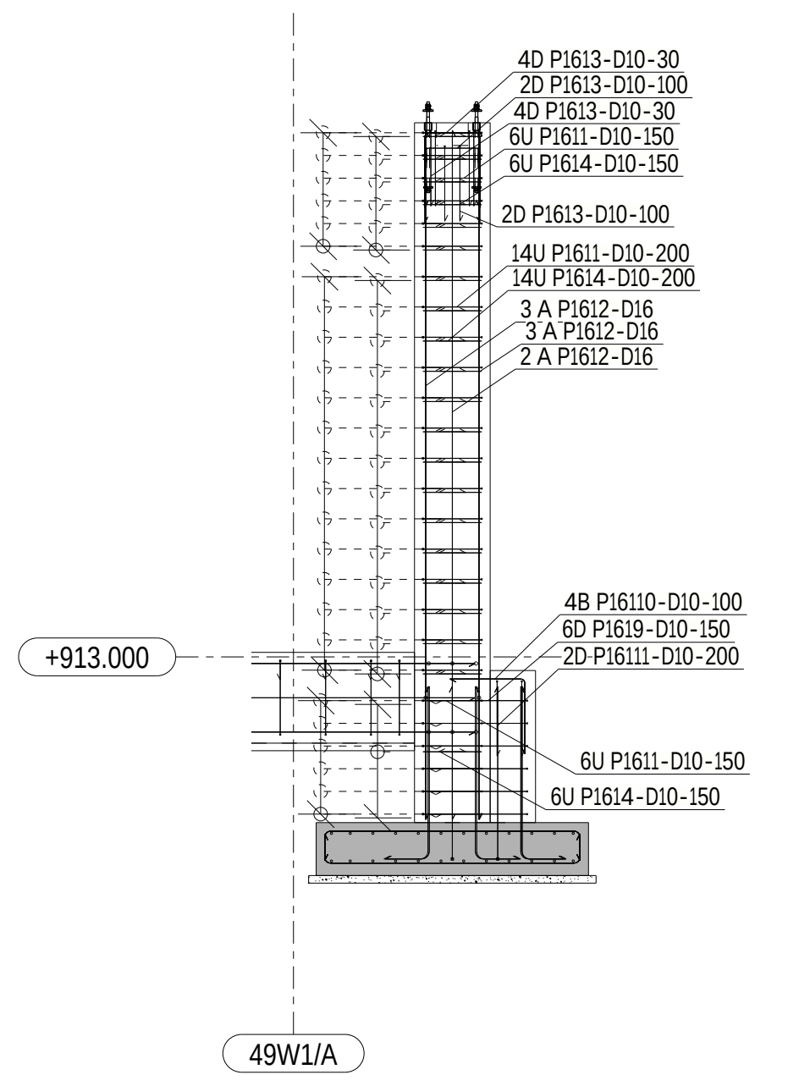
SECTION : B - B
(SCALE 1:50)


BEAMS AND SLAB ARE CAST SEPARATELY, FOAMED PLASTIC BETWEEN THE CASTS
F.G.L. +912.793
+913.000
F.G.L. +912.777
49W1/A

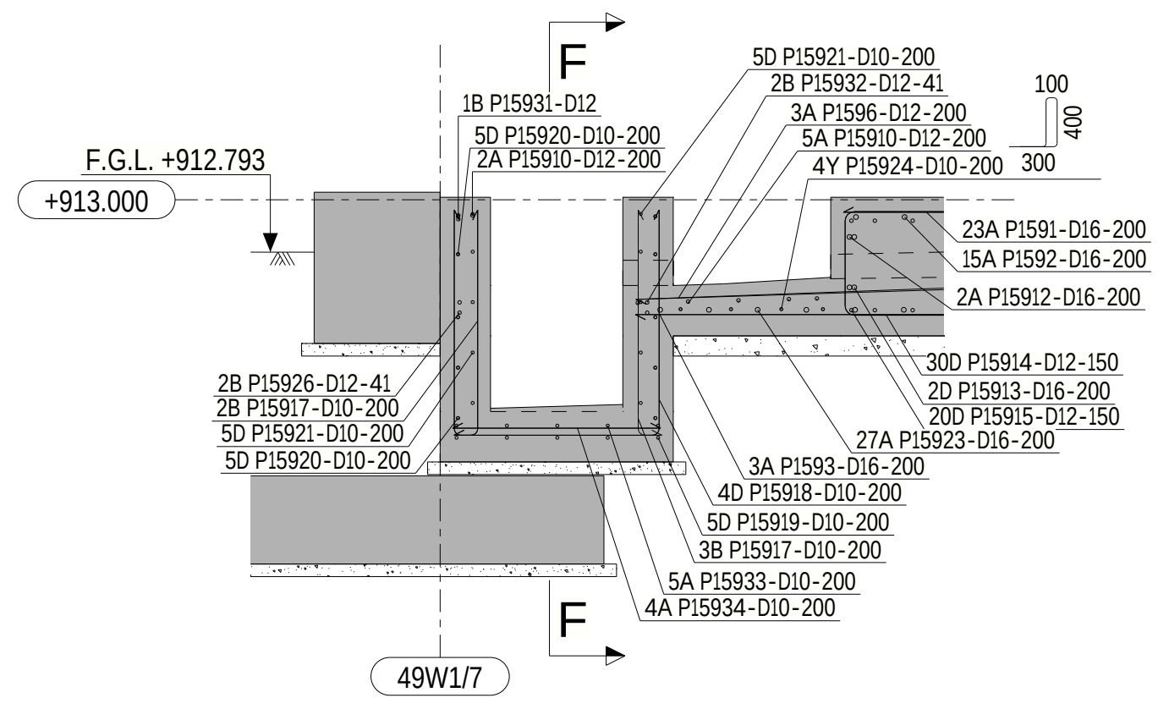
SECTION : C - C
(SCALE 1:30)


BEAMS AND SLAB ARE CAST SEPARATELY, FOAMED PLASTIC BETWEEN THE CASTS
F.G.L. +912.786
+913.000
F.G.L. +912.782
49W1/7
49W1/8

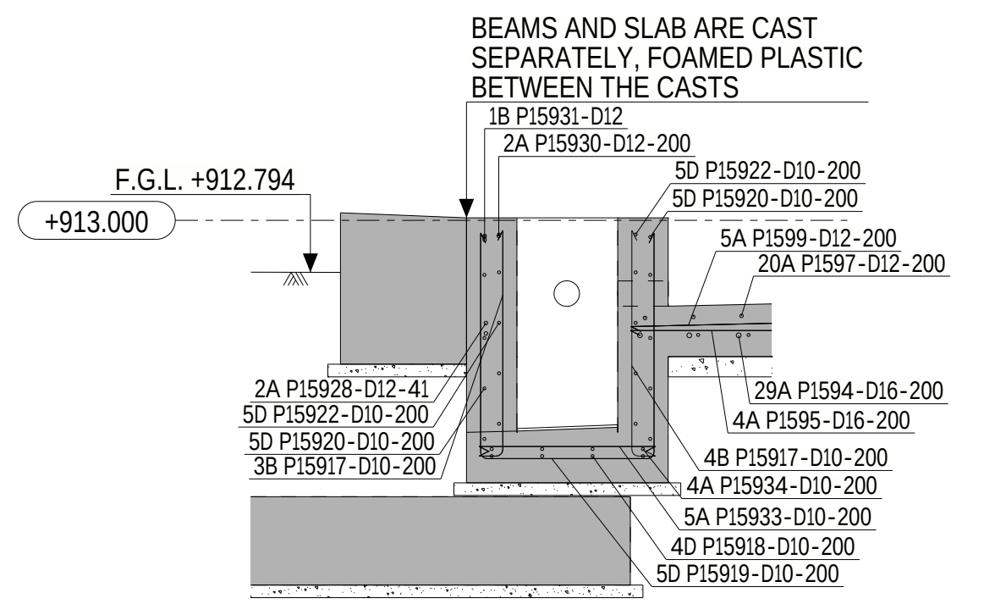
SECTION : D - D
(SCALE 1:30)


+913.000
49W1/A

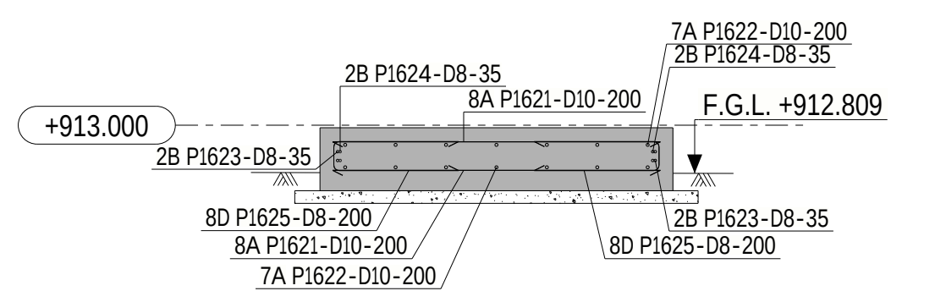
RC. DET. OF COLUMN P161-7 / P161-8
(SCALE 1:50)


F.G.L. +912.793
+913.000
F
F
49W1/7

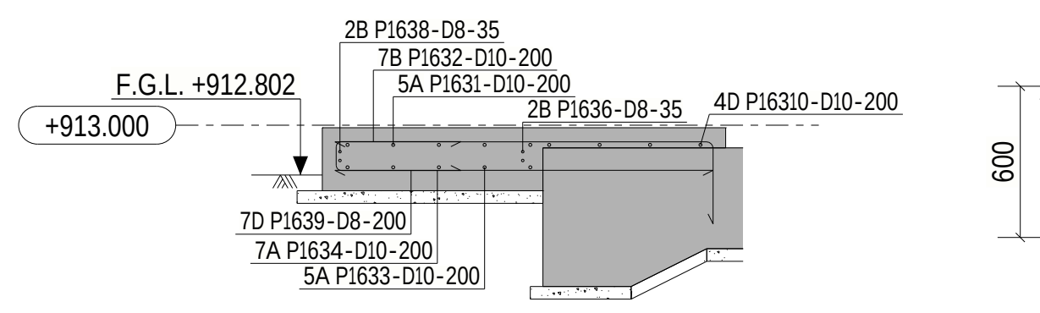
SECTION : E - E
(SCALE 1:30)


BEAMS AND SLAB ARE CAST SEPARATELY, FOAMED PLASTIC BETWEEN THE CASTS
F.G.L. +912.794
+913.000

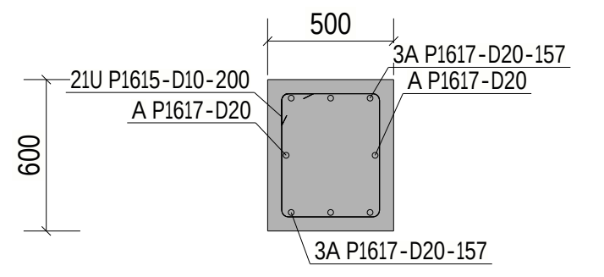
SECTION : F - F
(SCALE 1:30)


+913.000
F.G.L. +912.809

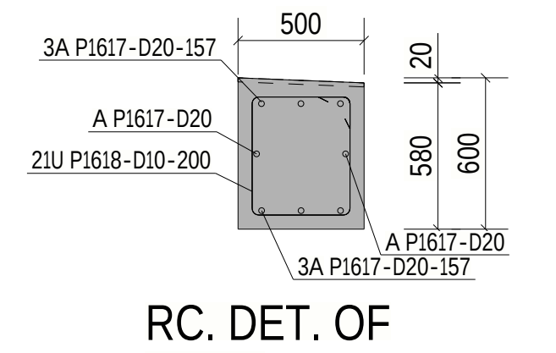
SECTION : G - G
(SCALE 1:30)


F.G.L. +912.802
+913.000

SECTION : H - H
(SCALE 1:30)



RC. DET. OF P1613 / P1614 / P1616
(SCALE 1:30)


500
580
600
3A P1617-D20-157
3A P1617-D20-157

RC. DET. OF P1615
(SCALE 1:30)

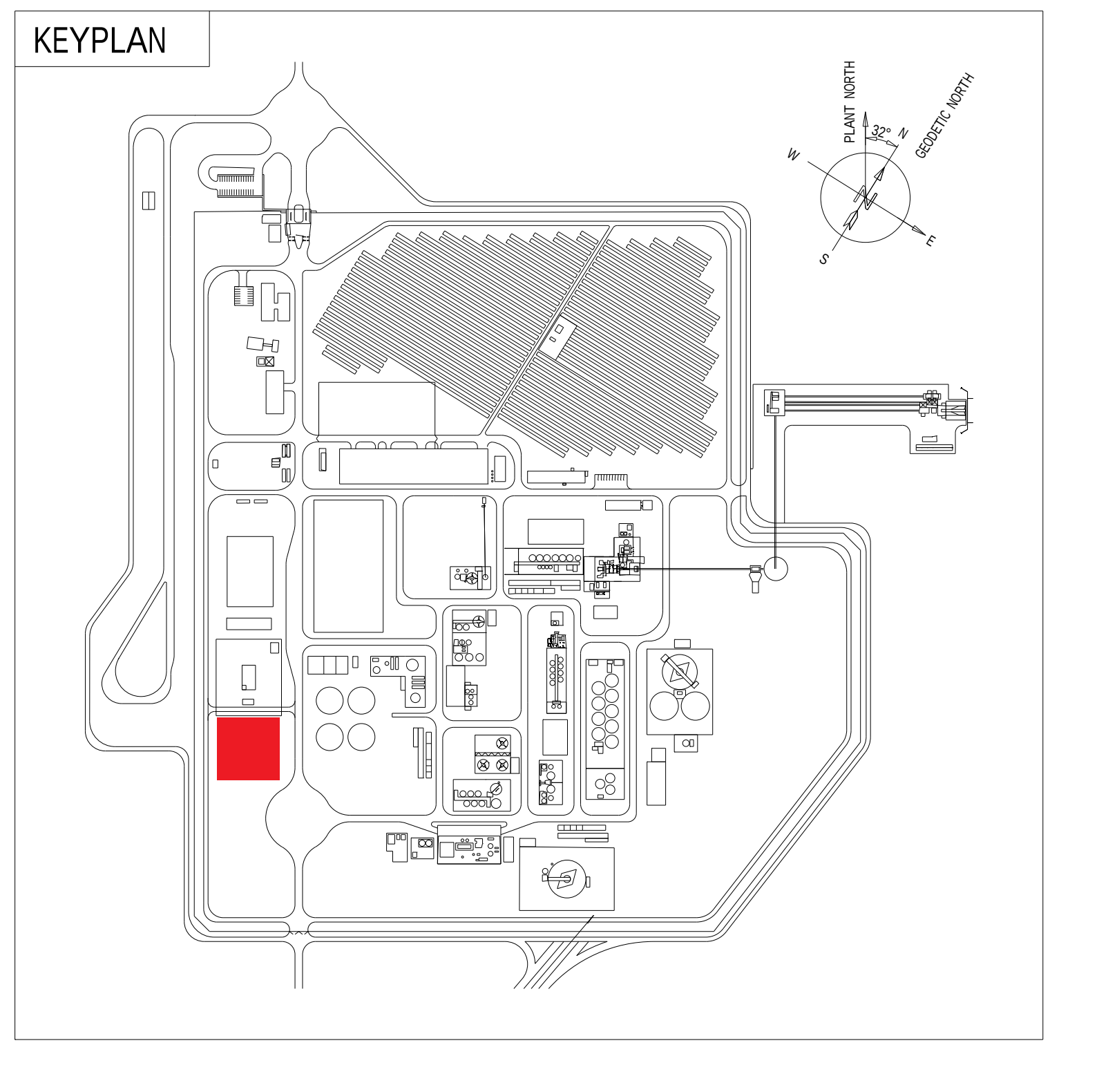
LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

KEYPLAN


PLANT NORTH
GEODETIC NORTH
32°
N

CONCRETE INFORMATION

Concretes used	C35	
Concrete cover 1	75±10mm	BOTTOM OF SLAB
Concrete cover 2	50±10mm	SIDES AND TOP OF SLAB, WALLS, BEAMS AND COLUMNS

REINFORCING BAR LIST

CAST UNIT CODE: P159 PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
A	P1591	23	B500B	16	2800		101.8	2800									
A	P1592	15	B500B	16	4350		103.1	4350									
A	P1593	3	B500B	16	5170		24.5	5175									
A	P1594	29	B500B	16	5900		270.3	5900									
A	P1595	4	B500B	16	5570		35.2	5575									
A	P1596	3	B500B	12	1420		3.8	1425									
A	P1597	43	B500B	12	2150		82.1	2150									
A	P1598	9	B500B	12	5900		47.1	5900									
14	AT GROUP	9	PCS	...	5880	3	...	5885									
A	P1599	5	B500B	12	5570		24.8	5576									
A	P15910	17	B500B	12	6170		93.2	6175									
A	P15911	15	B500B	12	2420		32.3	2425									
A	P15912	4	B500B	16	4320		27.3	4320									
D	P15913	4	B500B	16	4670		29.5	1009	2732	1009						80	
D	P15914	59	B500B	12	1360		71.5	500	425	500						60	
D	P15915	40	B500B	12	1300		46.5	500	369	500						60	
B	P15917	12	B500B	10	970		7.2	100	900				90			40	
D	P15918	4	B500B	10	2570		6.4	900	824	900						40	
D	P15919	5	B500B	10	2450		7.6	900	699	900						40	
D	P15920	10	B500B	10	2110		13.1	685	795	685						40	
D	P15921	10	B500B	10	790		4.9	85	670	85						40	
D	P15922	10	B500B	10	910		5.7	85	794	85						40	
A	P15923	27	B500B	16	6170		262.4	6174									
8	AT GROUP	27	PCS	...	6120	2	...	6126									
**	P15924	42	B500B	10	1400		36.5										
**	P15925	58	B500B	10	1440		51.8										
B	P15926	4	B500B	12	6750		24.0	6085	700				90			60	
B	P15927	2	B500B	12	6480		11.5	5810	700				90			60	
A	P15928	2	B500B	12	5790		10.3	5798									
A	P15930	4	B500B	12	5900		21.0	5900									
B	P15931	4	B500B	12	1370		4.9	700	700				90			60	
B	P15932	2	B500B	12	1670		3.0	700	1002				90			60	
A	P15933	5	B500B	10	700		2.2	699									
A	P15934	4	B500B	10	820		2.0	825									

REINFORCING BAR TOTAL WEIGHT [kg] 1467.3

REINFORCING BAR LIST

CAST UNIT CODE: P160 PCS: 4

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
D	P1601	96	B500B	12	2010		171.5	186	1700	186						60	
D	P1602	104	B500B	12	2040		188.4	200	1700	200						60	
B	P1603	38	B500B	16	1400		84.6	1150	300				90			80	

REINFORCING BAR TOTAL WEIGHT [kg] 444.5

REINFORCING BAR LIST

CAST UNIT CODE: P161 PCS: 9

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1611	104	B500B	10	1740		112.2	132	400	400	400	400				40	
A	P1612	32	B500B	16	4550		230.3	4555									
D	P1613	48	B500B	10	1320		39.4	500	376	500						40	
U	P1614	104	B500B	10	1350		86.9	132	301	301	301	301				40	
U	P1615	65	B500B	10	1940		78.1	132	500	400	500	400				40	
A	P1616	16	B500B	20	7070		279.6	7075									
A	P1617	16	B500B	20	6800		268.7	6800									
U	P1618	21	B500B	10	1900		24.7	132	480	400	480	400				40	
D	P1619	6	B500B	10	1650		6.1	650	400	650						40	
B	P16110	4	B500B	10	970		2.4	500	500				90			40	
D	P16111	2	B500B	10	1320		1.6	500	370	500						40	

REINFORCING BAR TOTAL WEIGHT [kg] 1130.1

REINFORCING BAR LIST

CAST UNIT CODE: P162 PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
A	P1621	16	B500B	10	1300		12.8	1300									
A	P1622	14	B500B	10	1500		13.0	1500									
B	P1623	4	B500B	8	1940		3.1	1460	505				90			32	
B	P1624	4	B500B	8	1730		2.7	1250	505				90			32	
D	P1625	16	B500B	8	1080		6.9	500	125	500						32	
D	P1626	14	B500B	8	1050		5.8	500	95	500						32	

REINFORCING BAR TOTAL WEIGHT [kg] 44.3

REINFORCING BAR LIST

CAST UNIT CODE: P163 PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
A	P1631	5	B500B	10	1300		4.0	1300									
B	P1632	7	B500B	10	1810		7.8	1506	332				90			40	
A	P1633	5	B500B	10	1320		4.1	1325									
A	P1634	7	B500B	10	1500		6.5	1500									
B	P1635	2	B500B	8	1880		1.5	1400	505				90			32	
B	P1636	2	B500B	8	1220		1.0	735	505				90			32	
D	P1637	10	B500B	8	1060		4.2	500	100	500						32	
B	P1638	2	B500B	8	1730		1.4	1250	505				90			32	
D	P1639	7	B500B	8	1080		3.0	500	125	500						32	
D	P16310	4	B500B	10	1920		4.8	332	1312	332						40	

REINFORCING BAR TOTAL WEIGHT [kg] 38.2

REINFORCING BAR LIST

CAST UNIT CODE: P164 PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
A	P1641	51	B500B	16	7820		630.8	7828									
A	P1642	184	B500B	16	12000		3488.6	12000									
A	P1643	23	B500B	16	4020		146.4	4029									
A	P1644	21	B500B	16	1900		63.1	1903									
A	P1645	282	B500B	16	6520		2909.1	6529									
A	P1646	3	B500B	16	3630		17.2	3632									
B	P1647	3	B500B	16	2900		13.8	1009	1940				90			80	
A	P1648	3	B500B	16	10350		49.1	10358									
B	P1649	3	B500B	16	6230		29.6	1009	5266				90			80	
A	P16410	3	B500B	16	1880		8.9	1886									
D	P16411	3	B500B	16	3770		17.9	1009	1835	1009						80	
A	P16412	18	B500B	16	3650		103.9	3655									
D	P16413	3	B500B	16	9690		46.0	1009	7760	1009						80	
A	P16414	3	B500B	16	3990		19.0	3999									
YJ	P16415	323	B500B	16	1930		985.2	800	282	425	500		27			60	
YJ	P16416	52	B500B	16	1890		155.3	800	277	390	500		27			60	
B	P16417	152	B500B	16	1450		350.4	1200	300				90			80	
A	P16418	62	B500B	16	2420		237.8	2428									
YJ	P16419	53	B500B	16	1880		157.6	800	277	390	500		27			80	
YJ	P16420	27	B500B	16	1920		82.0	800	282	425	500		27			80	
A	P16421	5	B500B	16	10430		82.4	10434									
A	P16422	5	B500B	16	1840		14.6	1849									
A	P16423	5	B500B	16	5170		40.9	5175									
A	P16424	5	B500B	16	1770		14.0	1777									
A	P16425	5	B500B	16	4050		32.0	4055									
B	P16426	6	B500B	16	1750		16.7	900	900				90			80	

REINFORCING BAR TOTAL WEIGHT [kg] 9712.5

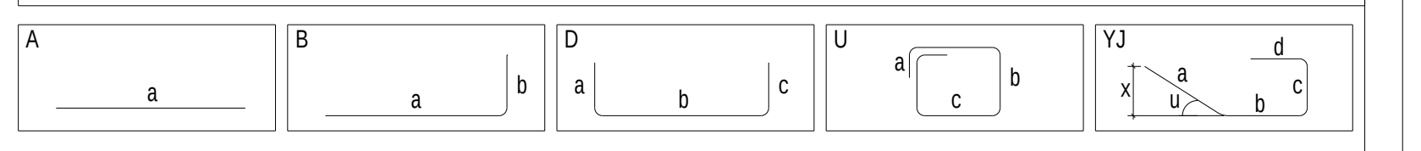
REINFORCING BAR LIST

CAST UNIT CODE: P165 PCS: 12

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	a	b	c	d	e	u	v	x	TD	COMMENT
U	P1651	192	B500B	12	2170		371.4	164	500	500	500	500				60	
A	P1652	152	B500B	16	2960		710.9	2960									
D	P1653	144	B500B	10	1420		126.7	500	472	500						40	
U	P1654	200	B500B	12	1670		297.3	164	500	247	500	247				60	
U	P1655	32	B500B	12	1690		48.2	164	379	379	379	379				60	

REINFORCING BAR TOTAL WEIGHT [kg] 1554.5

REINFORCEMENT BENDING SHAPES:


A
a
B
a
b
D
a
b
c
U
a
b
c
YJ
x
a
u
b
c
d

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-ME-GAD-1023	ELECTRICAL SUBSTATION DEMARCATION FENCE	MD-216-49W1-EG-CV-DED-1010	SUBSTATION AND TRANSFORMER FOUNDATION, DIMENSIONAL DRAWING
MD-216-49W1-EG-EL-GAD-V079-1001	1200KVA, 13.8/0.417 kV TRANSFORMER	MD-216-1000-EG-ST-CRT-1001	DESIGN CRITERIA
MD-216-49W1-EG-CV-CAL-1003	DESIGN CALCULATION FOR SUBSTATION FOUNDATION	MD-216-49W1-EG-CV-GAD-1003	GENERAL ARRANGEMENT DRAWING GL 1-10/A-E
MD-216-49W1-EG-EL-GAD-V079-1004	EQUIPMENT LAYOUT OF PREFAB BUILDING	MD-216-49W1-EG-CV-DED-1015	ANCHOR BOLT PLAN2

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	NikJan	MauNot	LuiPiz	SouCha	
C01	14.02.2022	ISSUED FOR CONSTRUCTION	NikJan	MauNot	LuiPiz	SouCha	
A02	27.01.2022	ISSUED FOR APPROVAL	KalKar	MauNot	NanWu	LuiPiz	
A01	22.12.2021	ISSUED FOR APPROVAL	KalKar	MauNot	NanWu	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to recieve it or be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_C_D_24892_021 — REVISION 3
OUTOTEC DRAWING NUMBER: 900707_GAU01_AG11_CTB01_49021 — DOCUMENT ID: OU100736747 — REVISION 3

DRAWN	KalKar	22.12.2021
CHECKED	MauNot	22.12.2021
ENGINEER	KalKar	22.12.2021
APPROVED	LuiPiz	22.12.2021
EMPLOYER		
SCALE	1:50	A0

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
SUBSTATION AND TRANSFORMER FOUNDATION
REINFORCEMENT DRAWING

DRAWING NUMBER: MD-216-49W1-EG-CV-DED-1021 — REV D01

Tekla Structures